17 **B**

$$\frac{1}{2}M\langle c^2\rangle = \frac{3}{2}kT$$

$$\therefore \frac{\frac{1}{2}M_P {c_P}^2}{\frac{1}{2}M_Q {c_Q}^2} = \frac{\frac{3}{2}kT_P}{\frac{3}{2}kT_Q}$$

$$\Rightarrow \frac{M_P}{M_Q} = \frac{T_P}{T_Q} \times \frac{{c_Q}^2}{{c_P}^2} = 2 \times \frac{1}{3^2} = \frac{2}{9}$$